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– User Experience Design (M. Sc.) –

UX and Climate Responsibility

Seminar Paper for the course

Critical UX Reflections

submitted by

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Statement about AI

AI was used for: Literature search, scope refinement, grammar checking, and structural guidance.

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1 Introduction

Digital services often feel "weightless". They are something you use daily without thinking much about the cost of them. They are services where a User Experience Designer tried to provide the best possible experience for their users. In their work process they have to make multiple choices, which in reality are not weightless at all. For example, a Netflix designer setting the default video quality to the highest possible one with autoplay enabled makes a choice that will be replicated for more than 325 million paid subscribers streaming over 95 billion hours in the first half of 2025 [20, 21]. A product manager deciding to implement an AI chatbot feature because it is a new trend, instead of a keyword search function, claims a share of data center infrastructure that consumes electricity from grids, still 58,9% powered by fossil fuels in 2024 in the United States [14].

While in 2024 data centers accounted for approximately 1.5% of global electricity consumption multiple factors make this ethically significant. Depending on different scenario cases this consumption share is predicted to rise to 2% - 4.4%, potentially tripling in a decade by 2035 [13]. Additionally data centres consume fresh water. In their 2025 environmental report Google claimed their total data centre water consumption in the year 2024 to be 29.5 billion litres of water growing yearly [11]. Lastly, this growth occurs against the backdrop of a climate crisis. In 2023 the Intergovernmental Panel on Climate Change warned that human-caused global warming has already reached 1.1 °C, triggering widespread and rapid changes across the biosphere that pose a direct threat to human well-being and planetary health [15]. Another report shows that of 34 tracked planetary vital signs, 22 are at record levels, many showing alarming trends [23]. The World Meteorological Organization confirmed that including 2025 the eleven past years were the 11 warmest years on record [30]. In addition, a 2026 report from the United Nations Universities Institute for Water, Environment and Health argues that for parts of the world the water crisis should rather be called water bankruptcy, since a crisis would suggest the possibility of turning back to normalcy [27]. These converging environmental pressures demonstrate the urgency of climate responsibility in all sectors, including digital UX design.

This paper argues that UX designers bear professional responsibility for environmental consequences that scale across billions of users, yet face barriers preventing ethical action. First, lack of transparency: tech companies treat energy consumption data as proprietary

information, making informed design decisions impossible. Second, structural pressure: designers are evaluated on engagement metrics (time-on-site, retention), not environmental impact. Proposing to disable autoplay or default to lower video quality means accepting reduced engagement scores and potential career consequences. Individual designers retain agency to advocate for sustainable choices within their professional sphere, but cannot overcome these barriers alone. The normative conclusion is therefore: designers should try to minimize environmental impact at every decision point and support regulatory interventions (mandatory energy disclosure, eco-friendly default standards) that make sustainable design professionally viable rather than professionally punished.

This paper proceeds in five parts. Section II establishes the materiality problem: the digital space is not weightless. Different UX design decisions have energy impacts that scale across billions of users. Section III analyzes this through ethical frameworks of climate justice and autonomy, demonstrating that current practices violate these principles by externalizing harm and denying both users and designers the information necessary for informed choice. Section IV examines the designer's professional dilemma in depth, demonstrating that even ethically-informed designers face severe structural constraints: business models reward engagement maximization regardless of environmental cost. Finally, Section V proposes normative recommendations.

2 UX and the Environment

Digital services are often perceived as immaterial. The "Cloud" suggests something ethereal and weightless. This perception obscures the physical reality. As Crawford argues, artificial intelligence and digital infrastructure are "profoundly material," requiring minerals extracted from the earth, electricity generated by power plants, and water consumed for cooling [8]. When designers make interface decisions, they are making claims on these material resources. This chapter highlights some UX decisions. Firstly the section highlights the power of defaults with the example of video streaming. Then it showcases three UX pattern examples that have eco-friendly alternatives. Lastly it talks about AI as a trend and its problems.

2.1 Defaults and Streaming Quality

When designing a video stream media player a designer must choose a default resolution. Default settings and digital nudging, which is the use of user-interface design elements to guide people's behavior in digital choice environments [29], are very powerful. In 2011 Jared Spool analyzed hundreds of Microsoft Word users settings files and found that less than 5% changed any of the 150+ settings at all [25]. Similarly in 2003 Eric J. Johnson and Daniel Goldstein showed that defaults have a big impact for even something very personal such as organ donation. In a country where opt-in is needed it resulted in 12% organ donors, whereas a country with opt-out, like Austria, being an organ donor had an acceptance rate of 99.98% [16]. Also with a direct connection to environmental footprint and responsibility, research showed that when a cargo transportation platform changed the default option from opt-in to opt-out for CO_2 offsetting, participation changed from 12% to stabilized 50% after 12 weeks. Additionally the firm didn't lose revenue [26].

Video quality defaults are a possible UX decision that has an environmental impact. This choice determines energy consumption for users, who will likely never adjust the setting or opt-out from the default. In 2018, researchers measured the same device on the same network and found that streaming in HD produced 130 grams of CO_2 per hour, while 4K produced 610 grams of CO_2 per hour which is nearly five times more. In 2024 a study by the Carbon Trust that found an entire European average of 55 grams per hour. This shows a general problem for these statistics. They are very context heavy and are dependent on energy grid, network and video quality, but the directional effect is consistent: higher resolution results in higher energy consumption [12, 6]. For a single user this seems insignificant but for Netflix's 325 million subscribers streaming 95 billion hours in the first half of 2025 alone [21, 20], even small per-user differences aggregate to large numbers. This shows individual design choices, multiplied across hundreds of millions of users who rely on default settings can result in significant environmental impact.

2.2 UX Pattern Decisions

Designers exercise environmental influence through everyday pattern decisions that dictate the "material weight" of digital interfaces. This is showcased by following examples:

- **Autoplay:** While implemented to maintain engagement, empirical research demonstrates that autoplayed preview videos can increase data transfer by up to 86% compared to static interfaces. This creates a large carbon footprint increase as platforms decode high-bit-rate video. [2].

- **Infinite Scroll:** Infinite scroll is classified as a "dark pattern" that exploits psychological vulnerabilities, similar to slot-machines, to bypass user intent and maximize attention capture[19]. By removing natural stopping points, this design forces the continuous loading of data centers and network resources for content that may never be actively consumed.

- **Media Assets:** A paper in 2023 analyzed two websites that were hosted with the same method. The difference was one site was a simple text-only website with little interaction. It generated 0.03 g of CO_2 per visit. The other site was more complex, designed with images, animated gifs, embedded videos and written articles. The second website generated 0.92 g of CO_2 per visit, which shows the big impact media assets can have [28].

These decisions again suffer from the scaling effect. One page visit seems negligible, but when a website gets millions of daily views or millions of social media users scroll longer than they actively wanted the environmental impact scales up.

2.3 AI as a Trend

AI reveals both the scale of impact and the opacity problem. It's estimated that training GPT-4 consumed 50 gigawatt-hours of energy. That is enough to power a city like San Francisco for three days. Training is a one-time cost. What matters more is inference, every time someone queries these models. Inference now accounts for 80 to 90% of AI's total computing power [22].

In August 2025, Google published the first comprehensive, first-party measurement of production AI serving energy consumption. Using a methodology that accounts for active AI accelerators, CPU and DRAM, idle machines, and data center overhead, they measured the median Gemini Apps text prompt at 0.24 watt-hours, emitting 0.03 grams of CO_2 equivalent and consuming about 0.26 milliliters of water. This is roughly the energy

impact of watching TV for about 9 seconds. Google claims this value has gotten about 44x more efficient over a 12 month period [9].

However, before Google's paper researchers argued that closed AI model providers are serving up a total black box . This is still mostly true and Google also didn't disclose data on complex prompts, image generations and video generation. This is a lack of transparency that is increasingly dangerous as we transition toward "reasoning models" which require up to 43 times more energy for simple tasks than traditional models. Additionally inference aggregates into a surge in infrastructure demand. After a decade of relative stability, data center electricity consumption has doubled since 2017. Projections suggest that by 2028, data centers will claim 12% of total U.S. electricity. Because AI requires "24-7" constant power, these facilities are increasingly turning to dirtier, carbon-intensive energy sources to meet immediate needs, resulting in a carbon intensity 48% higher than the U.S. average [22]. This showcases that AI as a trend has a significant environmental footprint and that the whole "Black box" dynamic brings further problems, when trying to make informed decisions.

These examples demonstrate that UX design decisions have measurable environmental consequences that scale across billions of users. This raises an ethical question: who bears responsibility for these consequences, and what principles should guide design decisions? Section 3 addresses this through frameworks of climate justice and autonomy.

3 Ethical Analysis

This section looks at climate justice and autonomy.

3.1 Climate Justice

In the context of digital and climate ethics, Distributive Justice plays a big part and concerns the "just distribution of burdens and benefits" associated with resource consumption and mitigation. It addresses the "Burden-Sharing Question": who has the moral responsibility to pay for the environmental costs of digital infrastructure? There is the Principles of Liability: Distributive justice suggests that "polluters" should bear costs in proportion

to their emissions. Furthermore, the "Beneficiary Pays Principle" argues that agents who have benefited from emissions-generating activities (such as tech companies and affluent users) have a duty to shoulder the resulting environmental burdens. Also the Global Commons is an important framework. This framework views the atmosphere and natural resources as a "global common" that should be distributed equally or based on subsistence needs rather than luxury consumption [5].

Current UX practices violate these principles. When designers default to 4K streaming or AI chatbots, they prioritize convenience for their users with high-speed internet and modern devices. The energy required to deliver this convenience is disproportionately generated in regions with carbon-intensive grids. Data centers currently happen to be clustered in places that have dirtier grids on average resulting in electricity that is 48% more carbon-intensive than the national average. For example Virginia hosts more data centers than any U.S. state [22]. These communities breathe the emissions. Additionally the resulting climate impacts, flooding, drought, displacement, disproportionately affect the Global South, which contributed least to cumulative emissions yet faces the most severe consequences [7, 24]. Lastly, climate change is a problem that is predicted into the future impacting generations that may yet have no possibilities in influencing the course. Northern users' convenience is traded for Southern climate risk, and local communities near data centers bear environmental costs for services they may never use. This constitutes distributive injustice. Those who benefit (convenience for privileged users) are not those who bear costs (climate-vulnerable populations, data center communities, future generations).

The distributive injustice of the digital sector is compounded by the normative dilemma of risk and uncertainty. As established in climate justice theory, the global "greenhouse gas budget" is not a fixed number but a variable dictated by our level of risk-aversion: choosing a 66% probability of meeting the 1.5°C target halves the permissible emissions compared to a 33% probability. According to the Precautionary Principle, when there is a threat of "massive loss" and the "mechanisms" of harm are understood, urgency is required regardless of precise prior probabilities. However, the industry's refusal to provide data prevents the fulfillment of the "knowledge condition" necessary to trigger this principle [5].

This leads directly to the "Black Box" problem of AI energy consumption. Because lead-

ing AI providers don't disclose exact consumption numbers, they deny the public the knowledge needed to identify these mechanisms. We are trapped in a state of forced uncertainty, where the "expected value" of our digital choices cannot be calculated, and the "massive loss" remains hidden behind a veil of corporate secrecy. This information asymmetry means that:

- Users cannot make informed choices about which services to use or how to use them
- Designers cannot make informed choices about which features to implement
- Regulators cannot set appropriate standards without accurate baselines
- Affected communities cannot contest decisions when impacts remain invisible

Especially those with power (tech companies) control both the infrastructure and information about its impacts. Those without power (users, designers, affected communities, regulators) are denied knowledge necessary to challenge the system. This is distributive injustice on a knowledge level: by hiding environmental costs, companies prevent accountability mechanisms that might redistribute burdens more fairly.

3.2 Autonomy

At its core, Autonomy is the ability to act in a self-determined way. Being able to govern one's life according to one's own reasons and motives rather than manipulative external influences. In the context of agency, an autonomous individual is one who initiates their own actions; any exercise of power over that individual is illegitimate unless they authorize it themselves. In UX design, autonomy is undermined by mechanisms that "make a mockery of our authority to determine our own actions" [3].

Dark patterns such as infinite scroll exploit psychological vulnerabilities like to bypass a user's reflective judgment and steal their attention. Similarly, autoplay can increase data transfer without the user ever making an authoritative judgment to consume that content. In these cases, the user's "authorization" is a mere formality; the real power to consume resources lies with an external "choice architecture" designed for engagement, not sustainability. Consider the typical streaming interaction: A user opens Netflix, sees an

interface with 4K autoplay enabled by default. As shown prior many users don't change default settings.

This creates an autonomy problem regardless of the user's underlying preferences. Some users might genuinely prefer 4K quality and would choose it even if fully informed about energy costs. For these users, the current default happens to align with their values. But other users prioritize environmental impact and would choose HD if aware of the difference. For these users, the default substitutes the designer's priorities for their own.

The autonomy violation is not that 4K is offered, but that the choice is made invisible through dark patterns (autoplay), information gaps (no disclosure of energy costs), and friction (settings buried in menus). Users cannot exercise authentic choice when they lack both information and accessible alternatives.

These ethical frameworks establish that current UX practices violate principles of distributive justice and autonomy. However, this raises a practical question: why do designers make these choices? Section 4 examines the structural constraints that create tension between ethical obligations and professional viability.

4 The Designers' Dilemma

The philosophical analysis thus far might suggest designers are moral failures, choosing profit over planet. This interpretation is too simple. Designers operate within structural constraints that severely limit individual agency.

4.1 Professional Pressure

Consider the possible professional reality: Designers often encounter significant challenges when integrating environmental sustainability into their work, primarily due to constraints imposed by organisational priorities and policies [4]. Designers are evaluated on engagement metrics, not sustainability. Proposing to disable autoplay or default to standard definition video quality might mean answering the question: "What impact will this have on engagement?" The honest answer "engagement will likely drop 10-15%" could end the conversation. The proposal is rejected because the company's survival depends on growth

metrics. Sustainability is a complex issue that involves multiple stakeholders, many not sharing the same level of enthusiasm for sustainability [4]. A designer who consistently prioritizes environmental impact over business metrics might face performance reviews questioning their alignment with company goals or might ultimately be replacement by someone less resistant. The gap between personal values and anticipated professional behavior reveals the structural nature of the problem. Designers retain agency. They make choices and bear some responsibility. But agency operates within severe constraints determined by business models that require infinite growth.

4.2 Interactive Exercise

To examine how ethical frameworks apply to real design decisions, I conducted an interactive exercise with 5 design students during this papers presentation. Participants were presented with a scenario: as a UX designer for a streaming platform, choose between two default configurations. Option A (Industry Standard): 4K default with autoplay enabled, yielding +30% engagement but 280g CO_2 per user per week. Option B (Planet-First): HD default with autoplay disabled, accepting -10% engagement but generating only 45g CO_2 per user per week. Voting showed 4 students choosing Option B and one choosing Option A. When asked "Who would choose still chose B if this was your actual job tomorrow, with your manager evaluating you on engagement metrics?", no students switched from B to A. When asking the student who chose A, why they chose like that, they answered pragmatically. As a designer they want the best quality possible with modern technologies to provide the best possible experience for the user. When asked "Who is responsible if you choose A and the planet suffers?", they responded: They don't think they are at fault for what energy is consumed and its rather the manufacturers responsibility. The limited sample size prevents generalizable conclusions, but the exercise illustrates a key tension: even designers with clear environmental values must navigate professional contexts where sustainability competes with engagement metrics. The one student who chose Option A articulated this tension: designers are hired to provide optimal user experience with available technology, and manufacturers bear responsibility for infrastructure efficiency. While this reflects legitimate professional reality, it underestimates designer responsibility for two reasons. First, design choices create demand that manufacturers must meet: responsibility is shared across the production chain. Second, "optimal expe-

rience" doesnt need to mean maximum resource consumption. Many users could prefer HD with faster loading if given informed choice. The structural pressure identified is real: designers who consistently prioritize environmental impact over business metrics risk career consequences. This tension between individual ethics and organizational incentives demonstrates why systemic solutions like regulatory standards and mandatory disclosure are necessary complements to individual responsibility.

4.3 Possible Objections

Having established that UX designers bear shared responsibility within structural constraints, three objections to this thesis warrant examination.

- **Scale:** Data centers represent only 1.5% of global emissions [13]. Why prioritize UX ethics over larger sectors? This misunderstands the argument's scope. The claim is not that UX causes climate catastrophe, but that professionals with disproportionate influence over a rapidly growing source bear responsibility commensurate with their power. Moreover, dismissing every sector below 5% of emissions would mean addressing none-climate change results from aggregating many "small" sources.

- **Efficiency:** Won't technological improvements solve this without changing UX patterns? Google documented a 44-fold emissions reduction per AI query over 12 months, demonstrating rapid progress [9]. However, total AI energy consumption continues growing [13]. This pattern is characteristic of Jevons Paradox: efficiency improvements reduce per-unit costs, which enables expanded consumption, resulting in higher total resource use. The 19th-century economist William Stanley Jevons observed this with coal. As engines became more efficient, coal consumption increased rather than decreased because efficiency made coal-powered applications economically viable in new contexts [1]. It has to be noted that recent scholarship questions whether Jevons Paradox applies straightforwardly to complex AI systems [18], but the pattern remains: without consumption constraints, efficiency enables expansion rather than reduction.

- **Business:** Won't mandatory disclosure and default changes harm businesses? The objection treats profit as inviolable while treating environmental harm as negotiable. This is the inverse of the justice framework established in Section III. When

regulation applies equally to competitors, it shifts industry norms rather than disadvantaging individual actors. This Objection also showcases that regulations must happen on a global scale.

Designers face genuine dilemmas between personal values and professional incentives. Individual ethics alone cannot overcome structural barriers that systematically reward engagement over sustainability. Section 5 therefore proposes systemic interventions that enable ethical design to become professionally viable.

5 What Should Be Done?

The analysis thus far reveals that individual designer responsibility is necessary but insufficient. Designers should advocate for sustainable choices within their professional sphere, but systemic change requires coordinated action at multiple levels. This section first identifies individual actions designers can take immediately, then proposes regulatory interventions necessary to make those actions professionally viable. The things Designers can do can be split into two layers. First the individual and professional actions:

- **Awareness:** You can't make ethical choices if you don't know the impact. Learn about the energy costs of the tools and patterns you use, teach others and support regulation.
- **Digital Restraint:** Just because Midjourney can generate 50 logo options doesn't mean you should request 50. Ask yourself: 'Do I need this, or is it just easy to do?'
- **Green UX or Planet Centered Design**
- **Reframe Sustainability as Performance**

In 2024 Kiourtis et al proposed best practices towards environmentally sustainable UX/UI and they showed that when applying these proposed best practices per view carbon footprint could be significantly reduced [17]. Similarily to this other Green UX or Planet Centered Design frameworks exist that should be known about. Even though these individual actions are important the analysis thus far reveals a structural problem: designers

bear responsibility for environmental consequences but are systematically denied the information necessary to act on that responsibility and act within systems where performance clashes with sustainability. Hence, I propose the following regulations.

- **Mandatory Disclosure Requirements:** Just as the European Union requires nutritional labeling on food products and energy ratings on appliances, digital services should display environmental impact metrics. Specifically, companies, applications and features should provide: (1) energy consumption per hour of use or per interaction (2) carbon emissions based on grid location where computable and (3) comparative information enabling informed choice.
- **Default Setting Standards:** Based on the power default settings hold, requiring eco friendly default settings can have a big impact.

Users can make informed consumption decisions, exercising their autonomy. Designers can make informed implementation decisions, fulfilling professional ethics obligations currently impossible without data. Regulators can establish appropriate standards based on empirical baselines rather than contested estimates. Precedent exists: the EU AI Act [10] mandates transparency in agentic AI systems that involve critical infrastructure; Carbon labeling represents a logical extension of existing digital transparency requirements. Critics might object that measurements vary by device, network, and context, making standardized labels impractical. However, appliance energy labels similarly provide ranges. The goal is not false precision but directional transparency: making environmental tradeoffs visible rather than invisible.

6 Conclusion

This paper has demonstrated that UX design decisions have measurable environmental consequences that scale across billions of users. The ethical analysis established that current practices violate principles of climate justice and autonomy. Distributive injustice occurs when users' convenience is subsidized by climate impacts on vulnerable populations. Autonomy violations occur when users are denied information necessary to make choices aligned with their own values. The "black box" problem, where companies do

not disclose energy consumption, creates an injustice that prevents both users and designers from making informed decisions. Designers operate within structural constraints that make individual ethical action insufficient. Performance evaluations based on engagement metrics rather than environmental impact create professional incentives that systematically punish sustainable design choices. While designers retain agency to advocate for eco-friendly alternatives, they cannot overcome business models that require growth maximization.

The normative conclusion is therefore twofold. First, designers bear professional responsibility to minimize environmental impact within their sphere of influence: being aware, learning about and advocating for sustainable alternatives, and supporting regulatory efforts. Second, systemic solutions are necessary to make ethical design professionally viable: mandatory energy disclosure enables informed choice, while regulatory default standards shift competitive dynamics to reward sustainability rather than punish it.

As digital services continue expanding and AI adoption accelerates, the question of who bears responsibility for environmental consequences will intensify. The framework developed here, combining individual professional responsibility with recognition of structural barriers requiring regulatory intervention, provides a path forward that avoids both naive individualism and fatalistic determinism. Designers must act within their sphere while advocating for systemic change that expands what ethical action can accomplish.

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